

Influence of titanium and gold platelet morphology on heat transfer in biomagnetic third-grade hybrid nanofluid flow through a vertical channel

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ABSTRACT

This study investigates the influence of heat radiation on entropy formation in an unstable pulsatile flow of a blood-based hybrid nanofluid through a permeable conduit. Due to its high viscosity, blood is modelled as a third-grade fluid. Gold and titanium nanoparticles, known for their extensive biomedical applications, are suspended in blood as the base fluid to form the hybrid nanofluid. The third-grade fluid has been injected from the left wall at the same velocity and withdrawn from the right wall. Through the use of perturbation, the nonlinear momentum and energy equations are reduced to nonlinear ordinary differential equations. The collocation method is utilized to solve these transformed equations. Graphic images and tables depict the effects of relevant parameters and emergent variables. The $Ti-Au$ hybrid nanofluid's steady velocity is higher than the Ti nanofluid. The Ti nanofluid's unsteady velocity is higher than the $Ti-Au$ hybrid nanofluid. The findings show that when the non-Newtonian parameter increases, the entropy generation falls. At the surface where y approaches zero, increasing thermal radiation and the Darcy number of the porous media results in a higher heat transfer rate.

1. Introduction

Heat transfer analysis has gained significant attention from researchers and engineers due to its potential to address the engineering challenges faced by various industries. The enhancement of thermal properties in common liquids like water, motor

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oil, and glycol is crucial for industrial and technological processes. To improve the thermal performance of fluids for industrial applications, more advanced liquids are being explored. The introduction of these new classes of liquids has successfully addressed production-related challenges, with Choi [1] pioneering work on metal microparticles, also known as nanofluids, and their oxide carrier fluids, being a key development. The study of thermal properties in fields like biotechnology, computer chips, and other electrical and mechanical systems has been made easier by these fluids. Enhancing base fluids' heat transfer efficiency is largely dependent on the special chemical and physical characteristics of nanoparticles.

In recent years, researchers have explored a new generation of nanofluids, called "hybrid" nanofluids (HNFs), which are created by dispersing nanoparticles or nanocomposite materials from mixing various metals to the base fluid. HNFs have shown to be more efficient and stable than single tiny fluids [2–8]. These hybrid nanofluids combine nanoparticles or nanocomposites of various metals suspended in a single fluid, optimizing the combination of diverse material properties. The primary objective of HNFs is to improve thermal performance by leveraging the unique characteristics of different nanoparticles. Chu et al. [9] investigated the amplification of thermal energy and solute particles using hybrid nanofluid flow across an extended convex surface when MHD, Joule heating, and activation energy are present.

The biological fluid present in living things that are impacted by magnetic fields is known as biomagnetic fluid. Blood is among the most well-known biomagnetic fluids. Because of its vital uses in bioengineering and the medical sciences, like minimizing bleeding during surgery, biomagnetic fluid research has gained increased attention in recent decades. Magnetic particles are also used in cancer treatment and magnetic resonance imaging (MRI) of particular human frame components. Gold nanoparticles are the key light-shedding substance in the biological sciences. The form, composition, minimal toxicity, and exceptional suitability for human utilization of gold nanoparticles have lately drawn the attention of researchers [10,11]. Titanium nanoparticles are extensively utilized in many facets of daily life. They are used in a variety of applications, including the food, pharmaceutical, fire retardant, insulator, surface protective coating, and lowering the toxicity of dyes. Titanium nanoparticles are used for increased efficacy, dose optimization, controlled and prolonged drug release and treatment with low toxicity [12–14].

The significance of non-Newtonian fluids in manufacturing processes and engineering such as beauty products, paints, pastes, plastic sheet formation, blood, paper making, as well as the prevalence of polymeric fluids in oil exploration, biomedical solutions, adhesives, and tars has recently caught the attention of a number of scholars. Differential, integral, and rate fluids are the three forms of fluids that are not Newtonian. Zhao et al. [15] investigated the mixed convective entropy-optimized magnetohydrodynamic (MHD) flow of a non-Newtonian fluid between two rotating discs. Chu et al. [16] analysed the radiative flow of a non-Newtonian nanoliquid over a stretched cylinder, considering the effects of bio-convection, Stefan blowing, and magnetism. A differential type fluid that has undergone successful examination is the third-grade fluid in several flow situations [17], and it is well established that it captures both normal stresses and effects that are not Newtonian, such shear thickening or thinning [18–24]. The blood, an important fluid, was determined to be of third grade throughout this assessment. Nazeer et al. [25] have studied the flow of a non-Newtonian fluid electroosmotically in a microchannel.

Recent interest in studying laminar flow and heat transfer in porous channels and pipes of various cross-sectional shapes stems from its critical role in industrial and biological processes. Examples include biological fluid flow in expanding or contracting capillaries, synchronized pulsations of porous diaphragms, processes like transpiration cooling, gaseous diffusion, the circulatory and respiratory systems, blood flow and dialysis in artificial kidneys, solid rocket propulsion, filtration and purification techniques, groundwater resource management, geological investigations, and applications in the petroleum sector [26–28]. Additional cases of magnetohydrodynamic pulsatile fluid flows with radiation in distinct geometries are discussed in [29–32].

Entropy generation is of interest to scientists and engineers because to its extensive application in chemical vapour deposition equipment, heat exchanger pumps, combustion, electronic cooling, porous media, solar collectors, and turbomachinery. Research on entropy formation is said to shed light on energy losses in industrial thermal systems and engineering. Many medical professionals and researchers are now interested in studying how entropy is generated in physiological systems in order to prevent unnecessary energy loss. Convection, radiation, evaporation, and conduction are the methods by which heat is transmitted inside the human body. Conduction and emission both lose energy when the surrounding heat is below 20 °C, and vaporized when the temperature is over 20 °C. Entropy estimates are therefore essential for defining these kinds of systems, which helps with the handling of such dire circumstances. Numerous analyses of entropy generation for various applications have been published [33–39].

The study gaps identified in the preceding literature reviews, we concluded that no research had been conducted to characterize the heat transport and entropy production properties of a titanium-gold/blood third-grade hybrid nanofluid in a vertical channel. As such, we sought to investigate A magnetic field's influence on the pulsatile flow of a Titanium-gold/blood base radiative hybrid nanofluid with viscous dissipation and heat source properties inside a porous channel. The work described in this article is immediately applicable to a variety of fields, including electronic equipment design, subsurface water resources, blood dialysis in an artificial kidney, biofluid networks, and heat dissipation in electronics, and so on. The non-dimensional flow field equations were solved using a semi-analytical technique. The present model's results are discussed via figures and tables.

2. Mathematical formulation

This study considers the flow of an incompressible, unsteady pulsatile biomagnetic Ti–Au/blood third-grade hybrid nanofluid in a permeable channel. The hybrid nanofluid, composed of gold and titanium nanoparticles suspended in blood, exhibits enhanced thermal and flow properties, crucial for biomedical applications. The flow occurs between two parallel walls: the left wall at $\hat{y} = 0$ is maintained at temperature T_1 , while the right wall at $\hat{y} = d$ is maintained at a higher temperature T_2 ($T_2 > T_1$). A steady and consistent

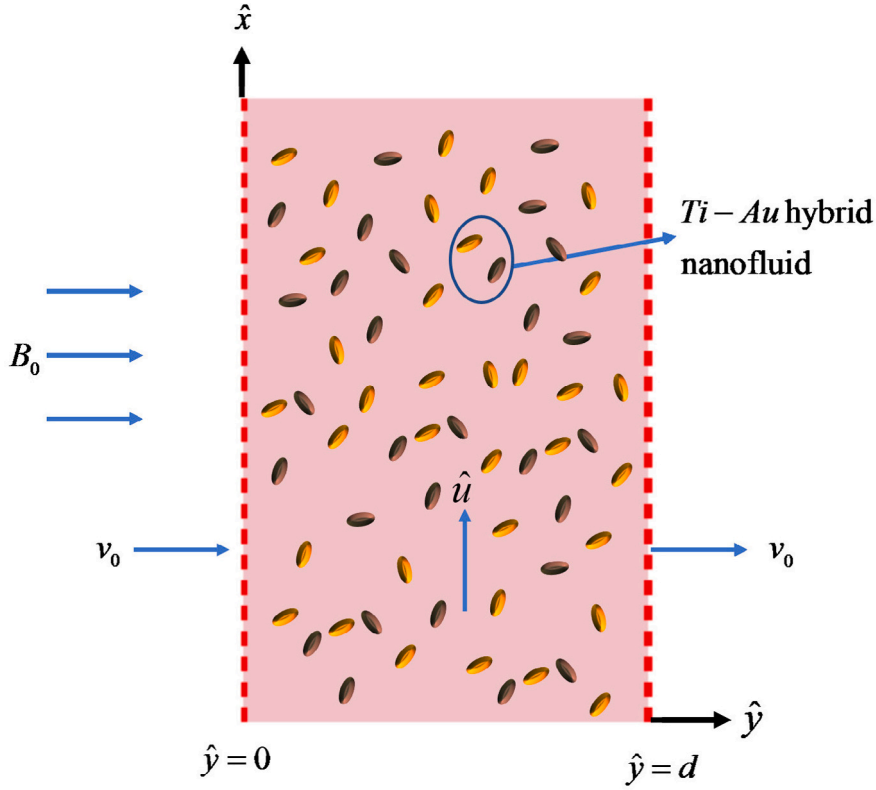


Fig. 1. A channel's coordinate system and physical model.

magnetic field interacts with an electrically conducting fluid, influencing its flow behaviour through magnetohydrodynamic (MHD) effects, as depicted in Fig. 1.

The flow is assumed to be fully developed, implying $\frac{\partial \hat{u}}{\partial \hat{x}} = 0$. This condition ensures the flow is uniform in the streamwise direction, leading to a solenoidal velocity field. Consequently, $\hat{v} = 0$, which means that the transverse velocity component $\hat{v}$ is constant across the channel. Since the no-slip boundary condition requires $\hat{v} = 0$ at both channel walls, it follows that $\hat{v} = 0$ throughout the domain. These assumptions significantly simplify the governing equations, focusing on the unsteady pulsatile behaviour of the primary velocity component $\hat{u}$ and the associated thermal effects in the hybrid nanofluid. Based on the given assumptions, the controlling flow equations are as follows [40,41]:

$$\frac{\partial \hat{u}}{\partial \hat{t}} + v_0 \frac{\partial \hat{u}}{\partial \hat{y}} = -\frac{1}{\rho_{hnf}} \frac{\partial \hat{p}}{\partial \hat{x}} + v_{hnf} \frac{\partial^2 \hat{u}}{\partial \hat{y}^2} + \frac{\alpha_1}{\rho_{hnf}} \frac{\partial^3 \hat{u}}{\partial \hat{t} \partial \hat{y}^2} + \frac{6\beta_3}{\rho_{hnf}} \left(\frac{\partial \hat{u}}{\partial \hat{y}} \right)^2 \frac{\partial^2 \hat{u}}{\partial \hat{y}^2} + g\beta_{hnf} (\hat{T} - T_1) - \frac{\sigma_{hnf} B_0^2}{\rho_{hnf}} \hat{u} - \frac{v_{hnf} \Phi}{k_0} \hat{u}, \quad (1)$$

$$\frac{\partial \hat{T}}{\partial \hat{t}} + v_0 \frac{\partial \hat{T}}{\partial \hat{y}} = \frac{k_{hnf}}{(\rho c_p)_{hnf}} \frac{\partial^2 \hat{T}}{\partial \hat{y}^2} - \frac{1}{(\rho c_p)_{hnf}} \frac{\partial q_r}{\partial \hat{y}} + \frac{Q_0}{(\rho c_p)_{hnf}} (\hat{T} - T_1), \quad (2)$$

with boundary conditions

$$\begin{aligned} \hat{u} &= 0, \hat{T} = T_1 \text{ at } \hat{y} = 0, \\ \hat{u} &= 0, \hat{T} = T_2 \text{ at } \hat{y} = d. \end{aligned} \quad (3)$$

where the coordinates of $\hat{x}, \hat{y}$, the velocity components $\hat{u}$ along the x -axis, and $\hat{v}$ along the y -axis, T_1 left wall temperature, $\hat{T}$ fluid temperature, T_2 right wall temperature, $\hat{p}$ pressure, v_0 mean suction velocity, $\hat{t}$ time, α_1 and β_3 are material constants, c_p specific heat capacity, k thermal conductivity, g acceleration due to gravity, ρ density, ν kinematic viscosity, μ dynamic viscosity, β coefficient of thermal expansion, σ electrical conductivity, f fluid, and hnf hybrid nanofluid.

The following are the nanofluid's physical properties:

$$\frac{\mu_{hnf}}{\mu_f} = \frac{1}{(1 - \phi_{Au})^{2.5} (1 - \phi_{Ti})^{2.5}},$$

$$\alpha_{hnf} = \frac{k_{hnf}}{(\rho C_p)_{hnf}},$$

$$\frac{\rho_{hnf}}{\rho_f} = (1 - \phi_{Ti}) \left((1 - \phi_{Au}) + \phi_{Au} \frac{\rho_{Au}}{\rho_f} \right) + \phi_{Ti} \frac{\rho_{Ti}}{\rho_f},$$

$$\frac{(\rho C_p)_{hnf}}{(\rho C_p)_f} = (1 - \phi_{Ti}) \left((1 - \phi_{Au}) + \phi_{Au} \frac{(\rho C_p)_{Au}}{(\rho C_p)_f} \right) + \phi_{Ti} \frac{(\rho C_p)_{Ti}}{(\rho C_p)_f},$$

$$\frac{\sigma_{hnf}}{\sigma_f} = \frac{(\sigma_{Ti} + (n-1)\sigma_{bf} - (n-1)\phi_{Ti}(\sigma_{bf} - \sigma_{Ti}))\sigma_{bf}}{(\sigma_{Ti} + (n-1)\sigma_{bf} + \phi_{Ti}(\sigma_{bf} - \sigma_{Ti}))\sigma_f},$$

$$\text{where } \sigma_{bf} = \frac{\sigma_f(\sigma_{Au} + (n-1)\sigma_f - (n-1)\phi_{Au}(\sigma_f - \sigma_{Au}))}{(\sigma_{Au} + (n-1)\sigma_f + \phi_{Au}(\sigma_f - \sigma_{Au}))},$$

$$\frac{(\rho\beta_t)_{hnf}}{(\rho\beta_t)_f} = (1 - \phi_{Ti}) \left((1 - \phi_{Au}) + \phi_{Au} \frac{(\rho\beta_t)_{Au}}{(\rho\beta_t)_f} \right) + \phi_{Ti} \frac{(\rho\beta_t)_{Ti}}{(\rho\beta_t)_f},$$

$$\frac{k_{hnf}}{k_{bf}} = \frac{(k_{Ti} + (n-1)k_{bf} - (n-1)\phi_{Ti}(k_{bf} - k_{Ti}))k_{bf}}{(k_{Ti} + (n-1)k_{bf} + \phi_{Ti}(k_{bf} - k_{Ti}))k_f},$$

$$\text{where } k_{bf} = \frac{k_f(k_{Au} + (n-1)k_f - (n-1)\phi_{Au}(k_f - k_{Au}))}{k_{Au} + (n-1)k_f + \phi_{Au}(k_f - k_{Au})}.$$

The gradient of unstable pressure is provided by

$$-\frac{1}{\rho_f} \frac{\partial \hat{p}}{\partial \hat{x}} = A [1 + \varepsilon \exp(i\omega \hat{t})] \quad (4)$$

where A is a known constant, $\varepsilon \ll 1$ is a suitably small positive quantity. The following are the non-dimensional quantities:

$$y = \frac{\hat{y}}{d}, \quad t = \omega \hat{t}, \quad u = \frac{\omega \hat{u}}{A}, \quad \theta = \frac{\hat{T} - T_1}{T_2 - T_1}, \quad P = \frac{\hat{P}}{A \rho_f d}, \quad Ec = \frac{A^2}{(c_p)_f \omega^2 (T_2 - T_1)},$$

$$M^2 = \frac{\sigma_f B_0^2 d^2}{\mu_f}, \quad H^2 = \frac{\omega d^2}{v_f}, \quad R = \frac{v_0 d}{v_f}, \quad Rd = \frac{4\sigma^* T_1^3}{k^* k_f}, \quad \chi = \frac{\beta_3 A^2}{\omega^2 d^2 \mu_f}, \quad (5)$$

$$Pr = \frac{\mu_f (c_p)_f}{k_f}, \quad Gr = \frac{g \beta_f (T_2 - T_1) \omega d^2}{A \mu_f}, \quad Da = \frac{k_0}{\Phi d^2}, \quad \delta = \frac{\alpha_1 \omega}{\mu_f}, \quad Q = \frac{Q_0 d^2}{(\rho c_p)_f v_f}.$$

Utilizing the non-dimensional quantities, Eqs. (1)–(3) turn into

$$H^2 \frac{\partial u}{\partial t} + R \frac{\partial u}{\partial y} = -\frac{1}{\rho_f} H^2 \frac{\partial P}{\partial x} + \frac{v_{hnf}}{v_f} \frac{\partial^2 u}{\partial y^2} + \frac{\delta}{\rho_f} \frac{\partial^3 u}{\partial t \partial y^2} + \frac{6\chi}{\rho_f} \left(\frac{\partial u}{\partial y} \right)^2 \frac{\partial^2 u}{\partial y^2}$$

$$+ \frac{\theta_{hnf}}{\beta_f} Gr \theta - \frac{\sigma_{hnf} M^2}{\sigma_f \rho_{hnf}} u - \frac{v_{hnf}}{v_f} \frac{1}{Da} u \quad (6)$$

$$\frac{(\rho c_p)_{hnf}}{(\rho c_p)_f} \left(H^2 \frac{\partial \theta}{\partial t} + R \frac{\partial \theta}{\partial y} \right) = \frac{1}{Pr} \left(\frac{k_{hnf}}{k_f} + \frac{4}{3} Rd \right) \frac{\partial^2 \theta}{\partial y^2} + \frac{\mu_{hnf}}{\mu_f} Ec \left(\frac{\partial u}{\partial y} \right)^2$$

$$+ 2Ec \chi \left(\frac{\partial u}{\partial y} \right)^4 + \frac{\sigma_{hnf} M^2 Ec}{\sigma_f} (u)^2 + Q \theta \quad (7)$$

with boundary conditions

$$u = 0, \theta = 0 \quad \text{at } y = 0,$$

$$u = 0, \theta = 1 \quad \text{at } y = 1. \quad (8)$$

where Pr is the Prandtl number, M is the magnetic parameter,

δ is the material parameter for third-grade fluid, Rd is the radiation parameter, Da is the Darcy number of the porous media, χ is the non-Newtonian coefficient, H is the frequency parameter, R is the cross flow Reynolds number, Q is the heat source parameter Gr is the thermal Grashof number, and Ec is the Eckert number.

3. Entropy generation

Enhancing heat transfer qualities in nanofluid flows leads to a reduction in entropy formation. A convection process involving a nanofluid channel flow, on the other hand, is intrinsically irreversible. Continuous entropy production is caused by non-equilibrium circumstances brought about by the momentum and energy exchange inside the nanofluid and at solid barriers. Heat transfer caused the formation of entropy in part towards limited temperature gradients and part by fluid friction, joule heating, and porous matrix.

The following provides the local volumetric rate of entropy formation:

$$SG = \frac{1}{T_2^2} \left(k_{hnf} + \frac{16\sigma^* T_1^3}{3k^*} \right) \left(\frac{\partial \hat{T}}{\partial \hat{y}} \right)^2 + \frac{\mu_{hnf}}{T_2} \left(\frac{\partial \hat{u}}{\partial \hat{y}} \right)^2 + \frac{2\beta_3}{T_2} \left(\frac{\partial \hat{u}}{\partial \hat{y}} \right)^4 + \frac{\sigma_{hnf} B_0^2}{T_2} (\hat{u})^2 + \frac{\mu_{hnf} \Phi}{T_2 k_0} (\hat{u})^2 \quad (9)$$

We may describe the entropy generation rate in non-dimensional form using Eq. (5).

$$N_G = \alpha \left(\frac{k_{hnf}}{k_f} + \frac{4}{3} Rd \right) \left(\frac{\partial \theta}{\partial y} \right)^2 + \frac{\mu_{hnf}}{\mu_f} Br \left(\frac{\partial u}{\partial y} \right)^2 + 2Br\chi \left(\frac{\partial u}{\partial y} \right)^4 + \frac{\sigma_{hnf} M^2 Br}{\sigma_f} (u)^2 + \frac{Br}{Da} \frac{\mu_{hnf}}{\mu_f} (u)^2 \quad (10)$$

where $Br = \frac{\mu_f A^2}{\omega^2 \Delta T k_f}$, is the Brinkmann number, $N_G = \frac{T_2 d^2 S_G}{\Delta T k_f}$ is the local entropy generation and $\alpha = \frac{\Delta T}{T_2}$ is the dimensionless temperature difference parameter.

4. Method of solution

In order to solve the corresponding nonlinear equations, a perturbation solution of the suggested form has been presented.

$$\begin{aligned} u(y, t) &= u_0(y) + \varepsilon u_1(y) e^{it}, \\ \theta(y, t) &= \theta_0(y) + \varepsilon \theta_1(y) e^{it}. \end{aligned} \quad (11)$$

where $\varepsilon \ll 1$ is a suitably chosen positive number. Substituting Eq. (11) into Eqs. (6)–(8), and approximating the coefficient of ε by neglecting the higher-order terms, we obtain:

$$A2 \left(\frac{d^2}{dy^2} u_0(y) \right) + 6\chi \left(\frac{d}{dy} u_0(y) \right)^2 \left(\frac{d^2}{dy^2} u_0(y) \right) - A1 R \left(\frac{d}{dy} u_0(y) \right) + H^2 + A3 Gr \theta_0(y) - A4 M^2 u_0(y) - \frac{A2 u_0(y)}{Da} = 0 \quad (12)$$

$$\begin{aligned} &+ A2 \left(\frac{d^2}{dy^2} u_1(y) \right) + i\delta \left(\frac{d^2}{dy^2} u_1(y) \right) - A1 R \left(\frac{d}{dy} u_1(y) \right) + 6\chi \left(\frac{d^2}{dy^2} u_1(y) \right) \left(\frac{d}{dy} u_0(y) \right)^2 \\ &+ 12\chi \left(\frac{d}{dy} u_1(y) \right) \left(\frac{d}{dy} u_0(y) \right) \left(\frac{d^2}{dy^2} u_0(y) \right) - iA1 H^2 u_1(y) + Gr A3 \theta_1(y) \\ &- A4 M^2 u_1(y) - \frac{A2}{Da} u_1(y) + H^2 = 0 \end{aligned} \quad (13)$$

$$\begin{aligned} &\frac{1}{Pr} \left(\frac{k_{hnf}}{k_f} + \frac{4}{3} Rd \right) \left(\frac{d^2}{dy^2} \theta_0(y) \right) - A5 R \left(\frac{d}{dy} \theta_0(y) \right) + Q \theta_0(y) + A4 M^2 Ecu_0(y)^2 \\ &+ 2\chi Ec \left(\frac{d}{dy} u_0(y) \right)^4 + A2 Ec \left(\frac{d}{dy} u_0(y) \right)^2 = 0 \end{aligned} \quad (14)$$

$$\begin{aligned} &\frac{1}{Pr} \left(\frac{k_{hnf}}{k_f} + \frac{4}{3} Rd \right) \left(\frac{d^2}{dy^2} \theta_1(y) \right) - A5 \left(\frac{d}{dy} \theta_1(y) \right) R + (-iA5 H^2 + Q) \theta_1(y) \\ &+ 2A4 Ec M^2 u_0(y) u_1(y) + \left(8\chi \left(\frac{d}{dy} u_0(y) \right)^2 + 2A2 \right) Ec \left(\frac{d}{dy} u_0(y) \right) \left(\frac{d}{dy} u_1(y) \right) = 0 \end{aligned} \quad (15)$$

with boundary conditions

$$\begin{aligned} u_0(1) &= 0, u_0(0) = 0, \theta_0(1) = 1, \theta_0(0) = 0, \\ u_1(0) &= 0, u_1(1) = 0, \theta_1(1) = 0, \theta_1(0) = 0. \end{aligned} \quad (16)$$

where $A1 = \frac{\rho_{hnf}}{\rho_f}$, $A2 = \frac{\mu_{hnf}}{\mu_f}$, $A3 = \frac{(\rho\beta)_{hnf}}{(\rho\beta)_f}$, $A4 = \frac{\sigma_{hnf}}{\sigma_f}$, $A5 = \frac{(\rho c_p)_{hnf}}{(\rho c_p)_f}$, $A6 = \frac{k_{hnf}}{k_f}$.

4.1. Solution process of collocation method

In the approximate solution for Eqs. (12)–(15), we introduced an appropriate velocity, and temperature trail functions are as follows:

$$\begin{aligned} u_0(y) &= a_1 y(1-y) + a_2 y(1-y)^2 + a_3 y(1-y)^3 + a_4 y(1-y)^4 \cdots a_n y(1-y)^n, \\ \theta_0(y) &= y + b_1 y(1-y) + b_2 y(1-y)^2 + b_3 y(1-y)^3 + b_4 y(1-y)^4 \cdots b_n y(1-y)^n, \\ u_1(y) &= c_1 y(1-y) + c_2 y(1-y)^2 + c_3 y(1-y)^3 + c_4 y(1-y)^4 \cdots c_n y(1-y)^n, \\ \theta_1(y) &= d_1 y(1-y) + d_2 y(1-y)^2 + d_3 y(1-y)^3 + d_4 y(1-y)^4 \cdots d_n y(1-y)^n, \end{aligned} \quad (17)$$

where $a's$, $b's$, $c's$ and $d's$ are finite number of undetermined coefficients. It is important for the trial functions Eq. (17) must satisfy the current model's boundary conditions Eq. (16).

When $u_0(y)$ and $\theta_0(y)$ are introduced into differential equations it yields residuals $R_1(a's, c's, y)$ and $R_2(a's, c's, y)$ as follows:

$$\begin{aligned}
 R_1(a_1, a_2, \dots, c_1, c_2, \dots, y) = & A2 \sum_{i=1}^N \left(-\frac{2a_i(1-y)^i}{1-y} + \frac{a_i y(1-y)^{i^2}}{(1-y)^2} - \frac{a_i y(1-y)^i}{(1-y)^2} \right) \\
 & + 6\chi \left(\sum_{i=1}^N \left(a_i(1-y)^i - \frac{a_i y(1-y)^i}{1-y} \right) \right)^2 \left(\sum_{i=1}^N \left(-\frac{2a_i(1-y)^i}{1-y} + \frac{a_i y(1-y)^{i^2}}{(1-y)^2} - \frac{a_i y(1-y)^i}{(1-y)^2} \right) \right) \\
 & - A1R \sum_{i=1}^N \left(a_i(1-y)^i - \frac{a_i y(1-y)^i}{1-y} \right) + H^2 \\
 & + A3Gr \left(y + \left(\sum_{i=1}^N c_i y(1-y)^i \right) \right) - A4M^2 \left(\sum_{i=1}^N a_i y(1-y)^i \right) - \frac{A2}{Da} \left(\sum_{i=1}^N a_i y(1-y)^i \right) \\
 R_2(a_1, a_2, \dots, c_1, c_2, \dots, y) = & \frac{1}{Pr} \left(\frac{k_{hbf}}{k_f} + \frac{4}{3} Rd \right) \left(\sum_{i=1}^N \left(-\frac{2c_i(1-y)^i}{1-y} + \frac{c_i y(1-y)^{i^2}}{(1-y)^2} - \frac{c_i y(1-y)^i}{(1-y)^2} \right) \right) \\
 & - A5R \left(1 + \sum_{i=1}^N \left(c_i(1-y)^i - \frac{c_i y(1-y)^i}{1-y} \right) \right) + Q \left(y + \left(\sum_{i=1}^N c_i y(1-y)^i \right) \right) \\
 & + A4M^2 Ec \left(\sum_{i=1}^N a_i y(1-y)^i \right)^2 + 2\chi Ec \left(\sum_{i=1}^N \left(a_i(1-y)^i - \frac{a_i y(1-y)^i}{1-y} \right) \right)^4 \\
 & + A2Ec \left(\sum_{i=1}^N \left(a_i(1-y)^i - \frac{a_i y(1-y)^i}{1-y} \right) \right)^2
 \end{aligned}$$

Now it is only a matter of altering the values of a 's and c 's to get an approximate solution for the issue in the interval $0 < y < 1$. As a result, the residual remains around zero over the period $0 < y < 1$. The essential assumption is that the residual between collocation points does not fluctuate substantially from zero. Let we take $N = 8$, $H = 2$, $R = 0.5$, $M = 0.5$, $Da = 0.2$, $\delta = 0.1$, $\chi = 0.1$, $Gr = 0.3$, $Pr = 21$, $Ec = 1$, $Q = 1$,

$$\begin{aligned}
 a_1 &= 1.426161163, a_2 = 0.03243866712, a_3 = -0.5533040444, a_4 = 1.414906698, \\
 a_5 &= -7.176162148, a_6 = 14.38830641, a_7 = -12.26932576, a_8 = 3.880014458, \\
 c_1 &= 3.055599098, c_2 = -7.111371621, c_3 = 8.172522103, c_4 = -2.525377321, \\
 c_5 &= -1.917255682, c_6 = 0.7239083231, c_7 = 1.417468316, c_8 = -0.8150915204,
 \end{aligned}$$

similarly

$$\begin{aligned}
 b_1 &= 0.63306528 - 0.13858612i, b_2 = 1.007103277 - 1.226864893i, \\
 b_3 &= -4.776669135 + 6.406946251i, b_4 = 21.28321269 - 27.30462815i, \\
 b_5 &= -56.83561561 + 67.97926346i, b_6 = 80.37626806 - 90.89791971i, \\
 b_7 &= -57.25164144 + 62.11592501i, b_8 = 16.32677002 - 17.17977765i, \\
 d_1 &= 1.240331696 - 1.508266461i, d_2 = -7.53722070 + 2.48952003i, \\
 d_3 &= 13.27354076 + 0.219719773i, d_4 = -9.994495919 - 2.96931077i, \\
 d_5 &= 8.64121481 - 11.15199604i, d_6 = -11.56260234 + 28.86215941i, \\
 d_7 &= 10.55098322 - 23.93158939i, d_8 = -3.72833566 + 7.21699017i.
 \end{aligned}$$

substituting $a's$, $b's$, $c's$ and $d's$ values in Eq. (17), we get

$$\begin{aligned}
 u_0(y) = & 1.426161163y(1-y) + 0.03243866712y(1-y)^2 - 0.5533040444y(1-y)^3 \\
 & + 1.414906698y(1-y)^4 - 7.176162148y(1-y)^5 + 14.38830641y(1-y)^6 \\
 & - 12.26932576y(1-y)^7 + 3.880014458y(1-y)^8,
 \end{aligned} \tag{18}$$

$$\begin{aligned}
 u_1(y) = & (0.6330653415 - 0.1385860971i)y(1-y) + (1.007102350 - 1.226865182i)y(1-y)^2 \\
 & + (-4.77666228 + 6.40694802i)y(1-y)^3 + (21.28318452 - 27.30463390i)y(1-y)^4 \\
 & + (-56.83554747 + 67.97927290i)y(1-y)^5 + (80.37617293 - 90.89792617i)y(1-y)^6 \\
 & + (-57.25157074 + 62.11592482i)y(1-y)^7 + (16.32674840 - 17.17977615i)y(1-y)^8,
 \end{aligned} \tag{19}$$

$$\begin{aligned}
 \theta_0(y) = & y + 3.055599098y(1-y) - 7.111371621y(1-y)^2 + 8.172522103y(1-y)^3 \\
 & - 2.525377321y(1-y)^4 - 1.917255682y(1-y)^5 + 0.7239083231y(1-y)^6 \\
 & + 1.417468316y(1-y)^7 - 0.8150915204y(1-y)^8,
 \end{aligned} \tag{20}$$

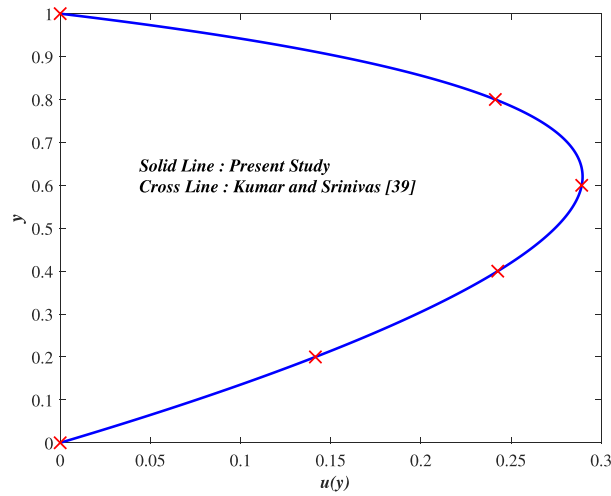


Fig. 2. The current study is compared to Kumar and Srinivas [39].

$$\begin{aligned} \theta_1(y) = & (1.240331676 - 1.508266497i)y(1-y) + (-7.537220572 + 2.489520393i)y(1-y)^2 \\ & + (13.27354022 + 0.219717296i)y(1-y)^3 - (9.994494308 + 2.969301248i)y(1-y)^4 \\ & + (8.641211540 - 11.15201772i)y(1-y)^5 + (-11.56259793 + 28.86218808i)y(1-y)^6 \\ & + (10.55097978 - 23.93160976i)y(1-y)^7 + (-3.728334427 + 7.216996139i)y(1-y)^8. \end{aligned} \quad (21)$$

$u(y, t)$ and $\theta(y, t)$ are obtained by substituting $u_0(y)$, $u_1(y)$, $\theta_0(y)$, and $\theta_1(y)$ in Eq. (11). Additionally, the Nusselt number without dimensions is provided by

$$Nu = - \left(\frac{k_{hnf}}{k_f} + \frac{4}{3} Rd \right) (\theta')_{y=0,1} = - \left(\frac{k_{hnf}}{k_f} + \frac{4}{3} Rd \right) (\theta'_0 + \varepsilon \theta'_1 e^{it})_{y=0,1} \quad (22)$$

5. Results and discussion

In this section of the research, the authors explain the findings of a schematic representation of significant physical variables in order to determine the quantitative variation versus specific essential parameters. In this regard, the authors have designed graphs illustrating steady and unsteady velocities, temperatures, entropy measures, and the heat transfer coefficient sequentially. In the present analysis $u_s(y)$, $u_t(y)$, $\theta_s(y)$, $\theta_t(y)$, NG represent steady velocity, unsteady velocity, steady temperature, unsteady temperature, entropy generation respectively. $\chi \in [0.1, 0.2, 0.3, 0.5]$, $M \in [0.0, 1.0, 2.0, 3.0]$, $Gr \in [0.0, 0.2, 0.4, 0.6]$, $R \in [0.0, 0.5, 1.0, 1.5]$, $Rd \in [0.0, 1.0, 2.0, 3.0]$, $Ec \in [0.2, 0.4, 0.6, 0.8]$, $Br \in [0.1, 0.2, 0.3, 0.4]$, $\alpha \in [0.1, 0.2, 0.3, 0.4]$, $t \in [0, \pi/3, \pi, 2\pi/3, 4\pi/3, 5\pi/3]$ refers to a set of significant parameter ranges chosen based on the convergence of the findings. Ti-Au Hybrid nanofluid case is shown by the dashed lines, whereas Ti nanofluid case is represented by solid lines. To evaluate the validity of the current model, Fig. 2 presents a comparison between the current results and those reported by Kumar and Srinivas [39], which demonstrates good agreement.

The consequences of steady velocities are portrayed in Fig. 3(a–d). The non-Newtonian coefficient χ influence on u is projected in Fig. 3(a). With an increase in the non-Newtonian coefficient, the steady velocity decreases because the heightened non-Newtonian parameter enhances viscoelasticity, thereby slowing the fluid's velocity. The modification of flow speed characteristics u in Fig. 3(b) helps us visualize the theoretical influence of the hydromagnetic component M . The graphic illustrates how the magnetic field and flow velocity have an inverse relationship, with the flow intensity being opposed by the magnetic field. This phenomenon arises naturally when a stronger magnetic field is applied in the opposite direction to the flow, creating a body force that resists the movement of fluid particles. In both cases, the velocity falls as the magnetic parameter rises. The Grashof number's effects on the velocity distribution is seen in Fig. 3(c). A laminar boundary layer is indicated by a high Gr , whereas a turbulent boundary layer is indicated by a low Gr . Improvements in the thermal Grashof number have resulted in higher thermal energy absorption by the fluid molecules, therefore relaxing the intermolecular connections between the fluid particles and enhancing the fluid's velocity profile and local heat transfer. The u -velocity figure becomes more trivial as the cross flow Reynolds number R increases, as seen in Fig. 3(d). In numerous fluid flow cases, the Reynolds number is instrumental in determining the flow characteristics. At lower Reynolds numbers, laminar flow is observed, whereas turbulent flow prevails at higher values. As the parameter R rises, the u -velocity patterns tend to moderate in both instances.

Fig. 4(a–e) illustrates the effects of non-Newtonian coefficient (χ), Magnetic parameter (M), thermal Grashof number (Gr) and cross-flow Reynolds number (R) and time (t) on unsteady velocity of hybrid nanofluid. The unsteady velocity of hybrid nanofluid decreases as the non-Newtonian parameter rises, as shown in Fig. 4(a). Increased non-Newtonian parameter values cause the fluid's

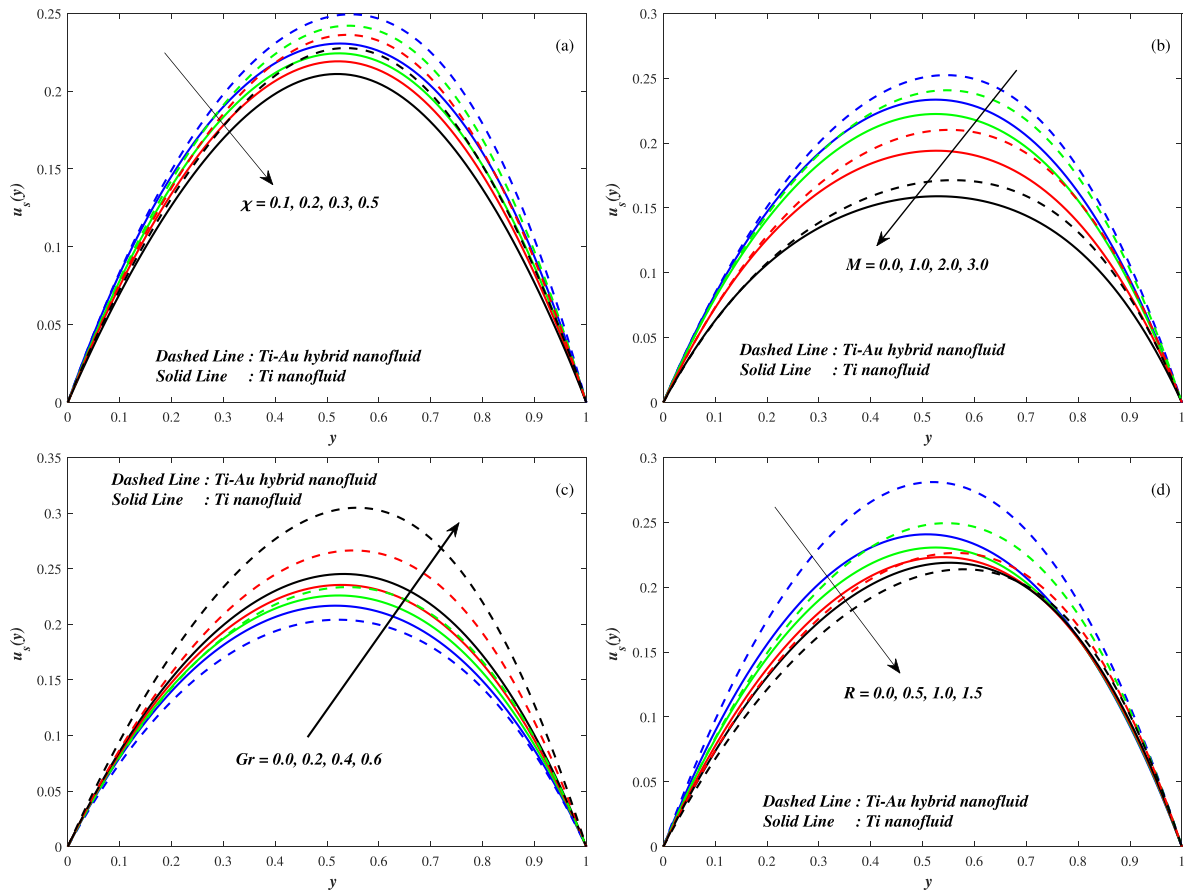


Fig. 3. u_s for uplifting values of χ , M , Gr and R for Ti-Au hybrid nanofluid and Ti nanofluid.

viscoelasticity to worsen and its velocity to decrease. It decelerates the blood velocity of the hybrid nanofluid when the applied magnetic field is increased, as seen in 4(b). This is because the magnetic field creates retarding forces (Lorentz forces), which cause this decline. These forces work in the opposite direction of the flow, slowing it down. As a result, the velocity decreases. Fig. 4(c) illustrates how the Grashof number affects the hybrid nanofluid's unstable velocity. The unsteady velocity decreases by intensifying the Grashof number. Fig. 4(d) notes that the velocity decreases as R increases. As the Reynolds numbers raise, then flow changes from laminar to turbulent. Fig. 4(e) shows the unsteady velocity profiles of the hybrid nanofluid for different t values. The velocity profiles in this diagram change with time t , with the velocity being most significant close to the centre of the channel.

Fig. 5(a–d) depicts the influence of M , Gr , Rd and Ec on steady temperature distribution $\theta_s(y)$. The effect of magnetic parameter M on the steady-state temperature is presented in Fig. 5(a). Applying a magnetic field can change the way fluid moves. Particles in the liquid form a chain that moves in the direction of an attractive field. They bump into each other during this time, making it hard for the fluid to flow. As a result, the fluid's speed slows down, and the nanofluid's temperature rises because the fluid's viscosity goes up. The variation in the temperature for different values of the thermal Grashof number is explained in Fig. 5(b) and found that there is a direct relationship between temperature and Gr . The buoyant forces brought on by temperature-induced density fluctuations are quantified by the thermal Grashof number in relation to the viscous forces in the boundary layer. A positive Gr value signifies that natural convection is cooling the hybrid nanofluid in the channel. Heat is conducted away from the channel into the nanofluid, which increases temperature and thereby enhances the buoyancy force. The impact of Rd on steady temperature is elucidated in Fig. 5(c). Raising the radiation parameter raises the hybrid nanofluid temperature because higher values of Rd expand the thermal barrier layer by adding additional energy to the working fluid. However, in the case of a heat source, the trend is the opposite. The physical features of temperature profiles of third grade fluid for greater values of Eckert number (Ec) are shown in Fig. 5(d). With increasing Ec values, temperature profiles appear to be rising. This is because the enthalpy factor decreases as Ec increases, causing particle kinetic energy to grow and, as a result, temperature to rise.

Fig. 6(a–f) present the variations in the unsteady temperature distribution θ_t for various parameters χ , M , Gr , Rd , Ec and t . These figures demonstrate that θ_t shows an oscillatory pattern, with its peak near both walls. Based on Fig. 6(a), it is evident that increasing χ results in a higher unsteady temperature. The time-dependent temperature of the blood increases with increasing M , as shown in Fig. 6(b). This is because The retarding forces of Lorentz operate against the flow, causing a decrease in the velocity of

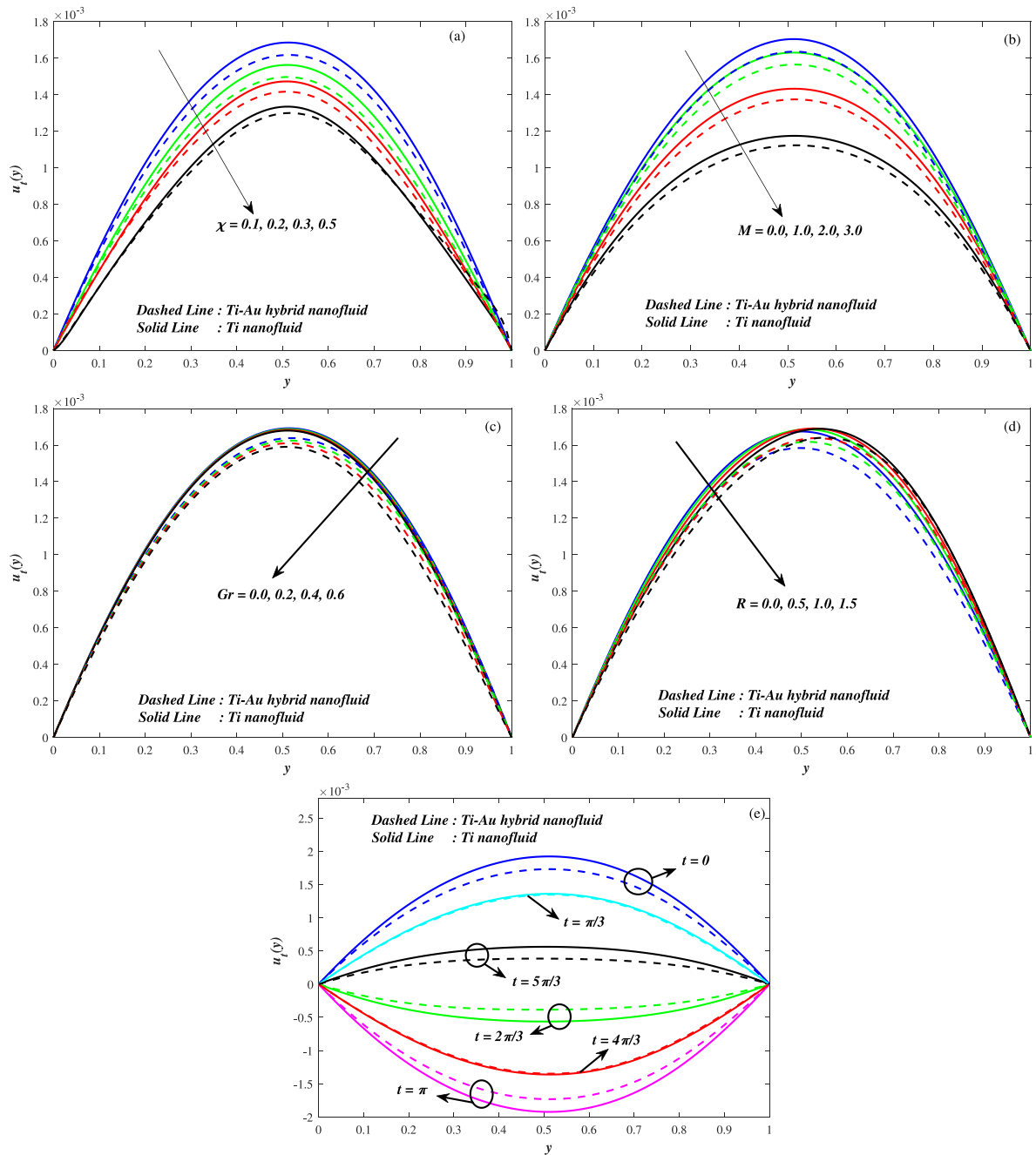


Fig. 4. u_t for uplifting values of χ , M , Gr , R and t for Ti-Au hybrid nanofluid and Ti nanofluid.

the blood in the boundary layer and a rise in its temperature. The role of the Grashof number in unsteady temperature variation is shown in Fig. 6(c). A laminar boundary layer is indicated by a greater thermo Grashof number, whereas a turbulent boundary layer is indicated by a lower value. Improvements in the thermal Grashof number have been found to result in higher thermal energy absorption by the fluid molecules, therefore relaxing the intermolecular connections between the fluid particles and enhancing the fluid's velocity profile and local heat transfer, which results in an improvement in temperature. Heat flows from the channel into the nanofluid, causing a rise in temperature and enhancing the buoyancy force.

Fig. 6(d) shows that, as Rd increases, there is a slight rise in temperature near the walls and a decline in the channel's centre. As the radiation parameter rises, this decrease may occur because the thermal boundary layer becomes thinner. As Fig. 6(e) illustrates, the unstable temperature shows an oscillatory behaviour with rising Ec , and the maximum temperature moves towards the boundary

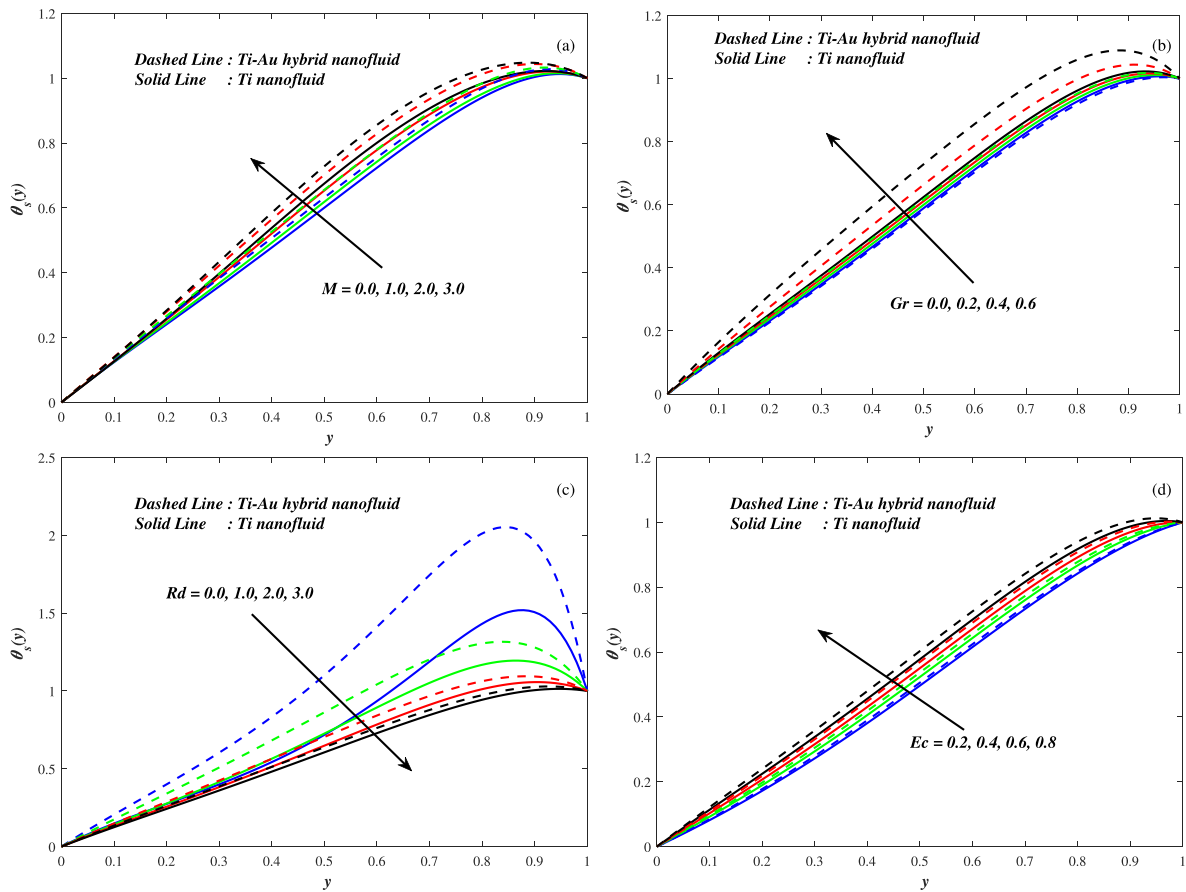


Fig. 5. θ_s for uplifting values of M , Gr , Rd and Ec for Ti–Au hybrid nanofluid and Ti nanofluid.

layers along the walls, showing an improvement in unsteady temperature profiles with higher Ec . It also shows that the unsteady temperature of the blood has an oscillatory nature. The temperature is highest along the walls compared to the centre of the channel, as seen in the same picture. Because Ec is connected with heat enthalpy and kinetic energy differences, by reducing the enthalpy factor, raising Ec increases particle kinetic energy, which raises the temperature. Changing the T causes the time-dependent temperature of the blood to oscillate; We can see this in Fig. 6(f).

The schematic diagrams of entropy generation have shown in Fig. 7(a–e), increasing the size of χ , M , Br , Rd and α . The influence of non-newtonian coefficient (χ) on entropy generation number (Nu) is presented in Fig. 7(a). χ shows indirect relation with the entropy generation. Compared to the hybrid nanofluid, the nanofluid has a lower entropy generation number. According to Fig. 6(b), entropy can be decreased by intensifying the effects of the magnetic field. Additionally, we observed that there is no energy loss in the channel's centre. In contrast to other flow zones, it therefore produced a significant amount of entropy along the right wall. It is evident from Fig. 7(c) that the Brikman number (Br) is directly related to the creation of entropy. According to the data shown in Fig. 7(c), the Ti nanofluid generates entropy at a lower rate than the Ti–Au hybrid nanofluid, which lowers energy consumption and boosts energy efficiency. The entropy generation has been plotted in Fig. 7(d) for various values of radiation parameter (Rd). The interesting point that should be mentioned is that, at higher values of Rd , the minimum value of entropy generation occurs for Ti nanofluid compared with the Ti–Au hybrid nanofluid. Fig. 7(e) illustrates the evolution of the entropy generation for different temperature difference parameter α . It is noted from the figure that entropy generation and temperature difference parameter are proportional to each other. Additionally, in the cases of the Ti and Ti–Au hybrid nanofluid, an ideal value is seen nearer the channel's right wall, and for the Ti–Au hybrid nanofluid, a maximum entropy production occurs.

Table 1 displays the thermophysical characteristics of the base fluid (blood) and the titanium and gold nanoparticles. Table 2 shows the heat transfer rate utilizing the Nusselt number at both the walls ($y = 0$ and $y = 1$) for the Steady and unsteady cases for various parameters. Table 2 shows that when Q and Ec increase, the Nusselt number at left wall $y = 0$ drops for both Steady and unsteady situations. Furthermore, the Nu increases as Q and Ec increases at $y = 1$ for the Steady case. The reverse trend is observed for increasing Q at $y = 1$ for an unsteady case. Table 3 is designed to check the validity and accuracy of the Nusselt number results by comparing the values of the different parameters of the channel wall at $y = 0$ and $y = 1$ for the stable and inconsistent cases by the bvp4C method. So we found that the current collocation method gave good results. Table 4 describes the deviations of Nusselt

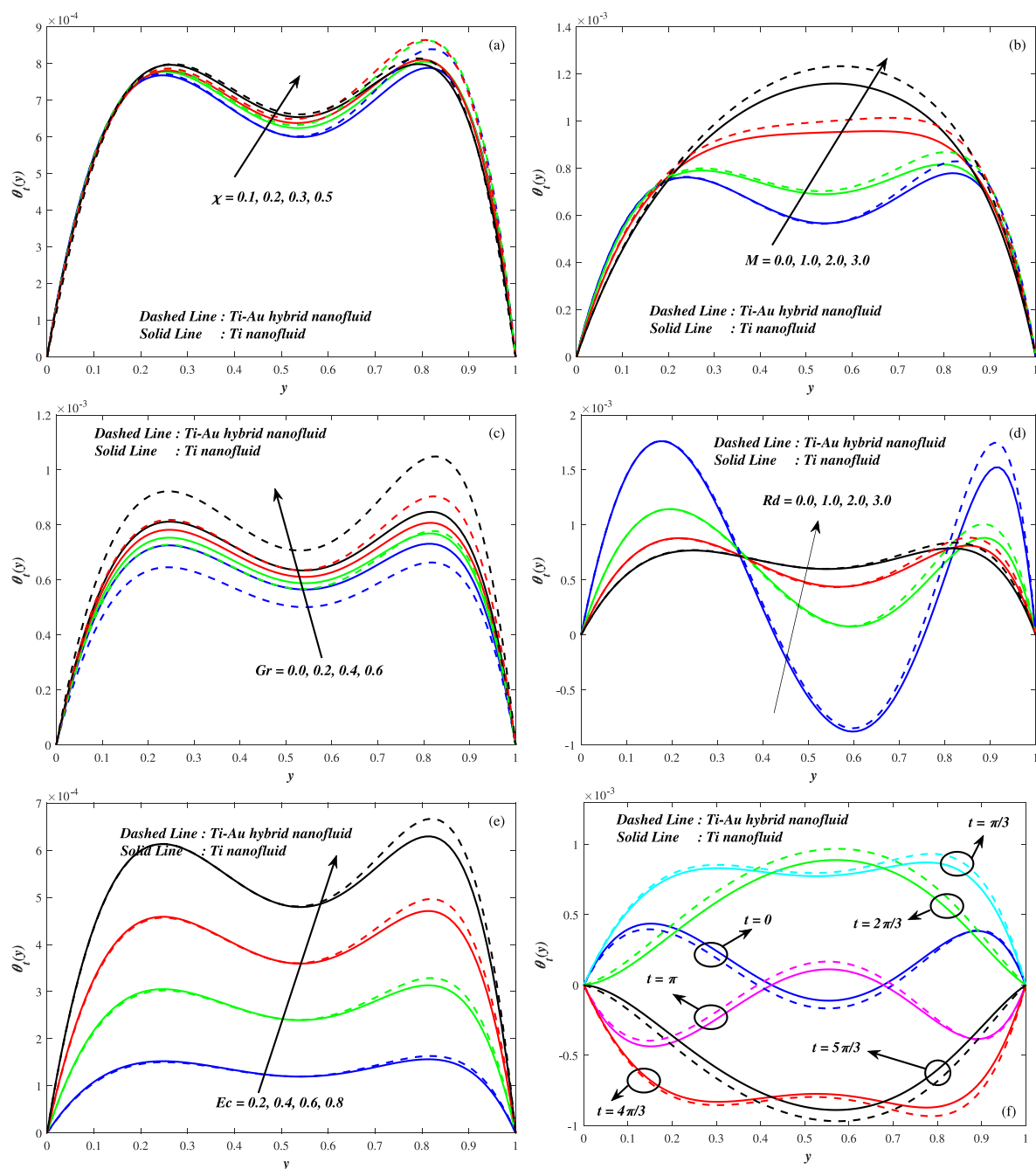


Fig. 6. θ_t for uplifting values of χ , M , Gr , Rd , Ec and t for Ti-Au hybrid nanofluid and Ti nanofluid.

Table 1
Thermophysical characteristics of the nanoparticles and base fluid.

	ρ (kg m ⁻³)	β_t (K ⁻¹)	k (W m ⁻¹ K ⁻¹)	σ (Ω ⁻¹ m ⁻¹)	c_p (J kg ⁻¹ K ⁻¹)
Blood	1050	0.18×10^{-5}	0.52	0.8	3617
Ti	4510	0.90×10^{-5}	20.9	2.5×10^6	540
Au	19282	14×10^{-6}	310	4.10×10^7	129

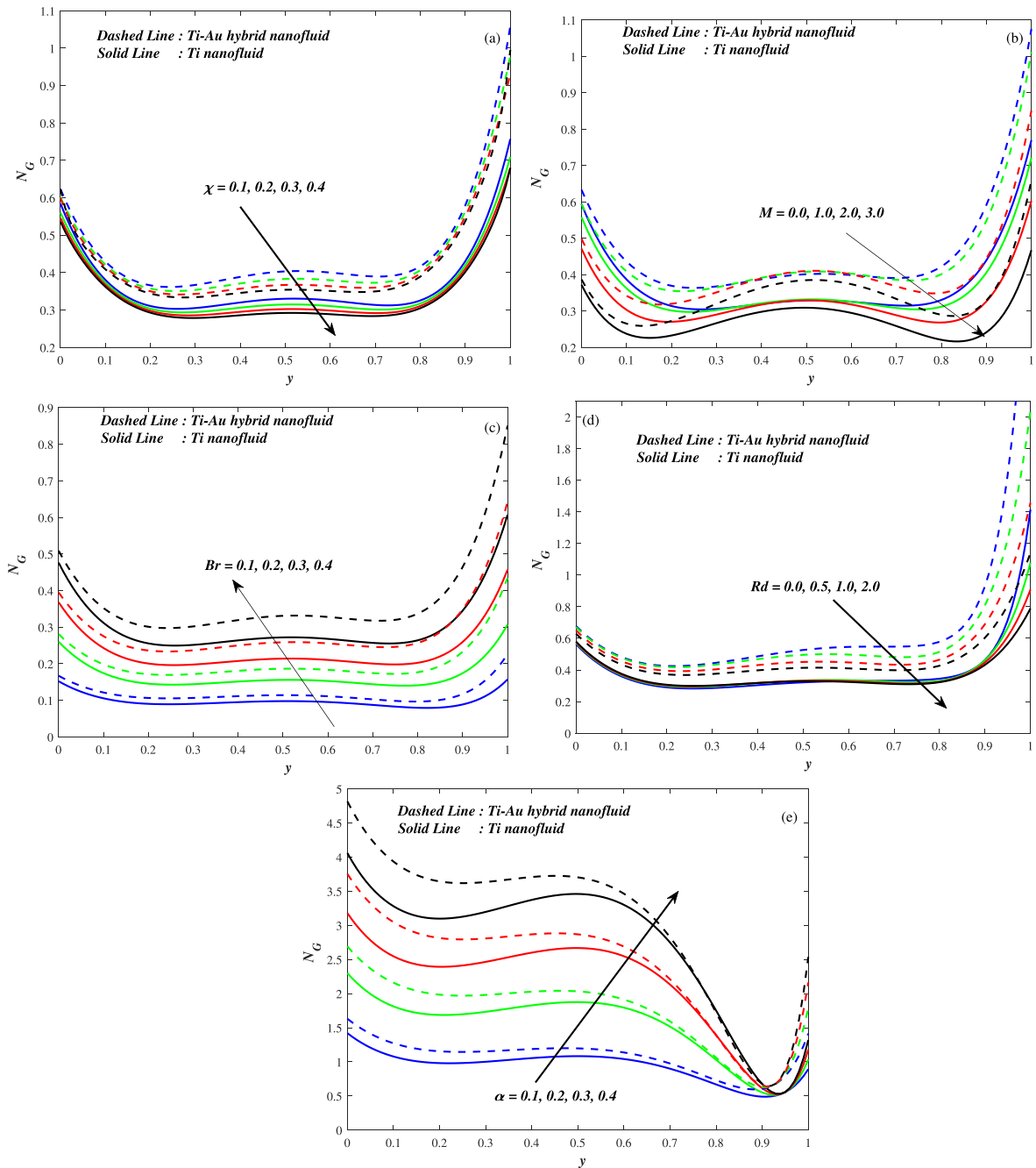


Fig. 7. N_G for uplifting values of χ , M , Br , Rd and α for Ti–Au hybrid nanofluid and Ti nanofluid.

number for different values of χ , δ , Gr , Da , M , R and H . From Table 4 it is clear that H , Da , Gr consuming a positive affiliation with the Nusselt number at $y = 0$ for steady and unsteady flows and a R , M , χ , consuming a negative affiliation with the Nu . δ keeps constant throughout the channel for steady flow. Nu accelerates for H , R , M , χ , δ at $y = 1$ for steady and unsteady flows and decelerates for Da and Gr . Table 5 exhibits the difference in Nusselt number for diverse values of $\phi_{Ti} + \phi_{Au}$, Q , Ec and Rd for steady and unsteady flows at the two walls of the channel $y = 0$ and $y = 1$. Declination in Nusselt number for $\phi_{Ti} + \phi_{Au}$ for steady flow at the wall $y = 0$ is noted whereas inclination for Rd , Ec , and $\phi_{Ti} + \phi_{Au}$ is observed. For unsteady flow at $y = 1$, an increment for $\phi_{Ti} + \phi_{Au}$ is noticed and a decrement for Ec and ϕ is witnessed. The radiation increases at the right wall, i.e., $y = 1$ for steady flow and a reverse trend is spotted for unsteady flow.

Table 2

Nu values for various values for Q and Ec with $H = 3, R = 1, M = 0.5, Da = 0.1, Rd = 3, \delta = \chi = Gr = \phi_{Ti} = \phi_{Au} = 0$ and $Pr = 21$.

Parameter	Value	Exact				Present			
		Steady		Unsteady		Steady		Unsteady	
		$y = 0$	$y = 1$	$y = 0$	$y = 1$	$y = 0$	$y = 1$	$y = 0$	$y = 1$
Q	0.2	-11.00991	6.28263	-0.14493	0.24798	-11.00990	6.28261	-0.14493	0.24798
	0.4	-11.26057	8.43534	-0.14530	0.24732	-11.26051	8.43536	-0.14530	0.24732
	0.6	-11.54244	10.84060	-0.14566	0.24659	-11.54246	10.84055	-0.14566	0.24660
	0.8	-11.86258	13.55590	-0.14601	0.24581	-11.86254	13.55590	-0.14601	0.24581
Ec	0.5	-6.41089	0.40638	-0.07318	0.12249	-6.41090	0.40637	-0.07317	0.12249
	0.6	-7.57480	3.65670	-0.08781	0.14699	-7.57480	3.65668	-0.08781	0.14699
	0.7	-8.73870	6.90701	-0.10245	0.17148	-8.73871	6.90700	-0.10244	0.17149
	0.8	-9.90261	10.15733	-0.11708	0.19598	-9.90261	10.15731	-0.11708	0.19598

Table 3

Nu Values for various of R, δ, χ, Gr with $H = 2, M = 0.5, Da = 0.1, Rd = 3, Pr = 21, Ec = 0.2, \phi_{Ti} = \phi_{Au} = 0$ and $Q = 0.2$.

Parameter values			bvp4C				Present method			
			Steady		Unsteady		Steady		Unsteady	
			$y = 0$	$y = 1$	$y = 0$	$y = 1$	$y = 0$	$y = 1$	$y = 0$	$y = 1$
$R = 0.5$	$\delta = \chi = 0.1$	$Gr = 0.3$	-2.24700	-9.71199	-0.00788	0.01140	-2.24700	-9.71201	-0.00788	0.01140
$R = 0.5$	$\delta = \chi = 0.1$	$Gr = 0$	-2.22874	-9.77047	-0.00777	0.01104	-2.22874	-9.77048	-0.00777	0.01104
$R = 0.5$	$\delta = \chi = 0$	$Gr = 0.3$	-2.25783	-9.69956	-0.00822	0.01190	-2.25783	-9.69956	-0.00822	0.01190
$R = 0$	$\delta = \chi = 0.1$	$Gr = 0.3$	-6.55874	-2.70601	-0.00957	0.00976	-6.55874	-2.70602	-0.00957	0.00976
$R = 0.5$	$\delta = \chi = 0$	$Gr = 0$	-2.23897	-9.76011	-0.00809	0.01149	-2.23896	-9.76012	-0.00809	0.01149
$R = 0$	$\delta = \chi = 0.1$	$Gr = 0$	-6.50880	-2.78120	-0.00930	0.00930	-6.50879	-2.78121	-0.00930	0.00930
$R = 0$	$\delta = \chi = 0$	$Gr = 0.3$	-6.56986	-2.69352	-0.01001	0.01022	-6.56986	-2.69352	-0.01001	0.01022
$R = 0$	$\delta = \chi = 0$	$Gr = 0$	-6.51898	-2.77102	-0.00970	0.00970	-6.51898	-2.77102	-0.00970	0.00970

Table 4

Nu values for various values of $\chi, \delta, Gr, Da, M, R$, and H with $\phi_{Ti} + \phi_{Au} = 0.05, Q = 0.2, Ec = 0.2, Rd = 3.0$.

χ	δ	Gr	Da	M	R	H	Steady		Unsteady	
							$y = 0$	$y = 1$	$y = 0$	$y = 1$
						1.5	2.29473	10.00146	0.00472	-0.00851
						2	2.89857	8.89969	0.01155	-0.01836
						2.5	4.06532	6.87975	0.02328	-0.03002
					0.6		5.54215	5.28312	0.05812	-0.04803
					0.7		4.79324	7.08220	0.05058	-0.04200
					0.8		4.15440	8.94608	0.04360	-0.03685
				1			6.12308	3.66737	0.05329	-0.04858
				2			5.49176	3.98949	0.03081	-0.03785
				3			4.94784	4.66741	0.02275	-0.03168
		0.25					7.16870	2.75099	0.08849	-0.06660
		0.3					7.80734	2.11701	0.10871	-0.07689
		0.35					8.34497	1.61411	0.12595	-0.08534
		0					5.91843	4.44135	0.05800	-0.05512
		0.1					6.07988	4.14853	0.06080	-0.05553
		0.2					6.24594	3.85566	0.06344	-0.05545
	0.2						6.41593	3.56508	0.06277	-0.05359
	0.3						6.41593	3.56508	0.05821	-0.05171
	0.4						6.41593	3.56508	0.05310	-0.04955
0.3							9.41171	4.21579	0.11656	0.00372
0.4							7.70969	7.27152	-0.02342	0.10738
0.5							4.64366	10.67630	-0.10173	0.14193

6. Conclusion

The analysis aims to enhance heat transfer in the pulsatile flow of a biomagnetic non-Newtonian hybrid nanofluid within a vertical channel. The perturbation method is applied to manage the coupled non-linear PDEs, and the effects of various parameters are studied in the context of two specific nanofluid cases: Ti–Au/blood hybrid nanofluid and Ti/blood nanofluid. The subject under consideration is critical in the biological arena. We consider the third-grade blood fluid as a base fluid. Gold and titanium nanoparticles are effective medication carriers because they can enclose significant amounts of bioactive factors. Because of this, we can use it as targeted drug delivery. The following are the key outcomes of this investigation:

- Steady and unsteady velocities decreases with increasing the values of χ M and R .

Table 5Nu values for various values of $\phi_{Ti} + \phi_{Au}$, Q , Ec , Rd with $H = 3$, $R = 0.5$, $M = 0.5$, $Da = 0.2$, $\delta = 0.1$, $\chi = 0.1$ and $Gr = 0.3$.

$\phi_{Ti} + \phi_{Au}$	Q	Ec	Rd	Steady		Unsteady	
				$y = 0$	$y = 1$	$y = 0$	$y = 1$
			1.5	4.496972	1.759283	0.058418	−0.05136
			2	5.128787	2.347439	0.061454	−0.05251
			2.5	5.769166	2.94996	0.063964	−0.0537
		0.05		3.023389	8.981937	0.015749	−0.0134
		0.1		4.127764	7.209428	0.031993	−0.027
		0.15		5.258155	5.404241	0.048768	−0.04086
	0.3			6.65523	2.642859	0.066811	−0.05508
	0.4			6.917787	1.659652	0.067602	−0.05522
	0.5			7.20701	0.607146	0.068392	−0.05534
0				7.174015	2.812037	0.135532	−0.11157
0.02				6.770751	3.223509	0.10059	−0.08143
0.04				6.504824	3.480452	0.075572	−0.06177

- M , Gr and Ec are helpful in enhancing the steady and unsteady temperatures of both hybrid nanofluid and nanofluid.
- Radiation decreases for steady and increases for the unsteady temperature of the nanofluid.
- χ , M , and Rd are constructive to enrich the thermal boundary thickness, so entropy generation minimizes with the rise of these parameters.
- Entropy generation and temperature difference parameter are proportional to each other, and a maximum entropy generation occurs for Ti–Au hybrid nanofluid.

CRedit authorship contribution statement

S. Suneetha: Conceptualization. **S.R.R. Reddy:** Data curation. **Maduru Lakshmi Rupa:** Data curation, Formal analysis. **T. Mahesh Kumar:** Investigation, Supervision. **Mohammed Jameel:** Resources, Writing – review & editing. **Taoufik Saidani:** Formal analysis, Visualization. **Lamia Abu El Maati:** Investigation, Methodology, Resources. **Dilsora Abduvalieva:** Data curation, Formal analysis, Writing – review & editing. **M. Ijaz Khan:** Formal analysis, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

References

- [1] S.U.S. Choi, J.A. Eastman, Enhancing thermal conductivity of fluids with nanoparticles, 1995.
- [2] A. Venkateswarlu, S. Suneetha, M.J. Babu, J.G. Kumar, C.S.K. Raju, Q. Al-Mdallal, Significance of magnetic field and chemical reaction on the natural convective flow of hybrid nanofluid by a sphere with viscous dissipation: A statistical approach, *Nonlinear Eng.* 10 (1) (2021) 563–573.
- [3] L. Wahidunnisa, S. Suneetha, S. Reddy, P.B.A. Reddy, Comparative study on electromagnetohydrodynamic single-wall carbon nanotube–water dusty nanofluid in the presence of radiation and ohmic heating, *Proc. Inst. Mech. Eng. E* 235 (4) (2021) 950–958.
- [4] A. Hamid, A. Hashim Hafeez, M. Khan, A.S. Alshomrani, M. Alghamdi, Heat transport features of magnetic water–graphene oxide nanofluid flow with thermal radiation: Stability test, *Eur. J. Mech. - B* 76 (2019) 434–441.
- [5] H.T. Basha, R. Sivaraj, I.L. Animasaun, Stability analysis on ag-mgo/water hybrid nanofluid flow over an extending/contracting rigid wedge and stagnation point, *Comput. Therm. Sci.* 12 (6) (2020) 491–508.
- [6] G. Sulochana, C.V. Prasad, S.K. Bhatti, V.V. Madhav, K.K. Saxena, M.I. Khan, Z. Aloui, C. Prakash, M.I. Khan, Impact of multi-walled carbon nanotubes (MWCNTs) on hybrid biodiesel blends for cleaner combustion in CI engines, *Energy* 303 (2024) 131911.
- [7] S. Jakeer, P.B.A. Reddy, A.M. Rashad, H.A. Nabwey, Impact of heated obstacle position on magneto-hybrid nanofluid flow in a lid-driven porous cavity with Cattaneo-Christov heat flux pattern, *Alex. Eng. J.* 60 (1) (2021) 821–835.

- [8] S.R.R. Reddy, P.B.A. Reddy, Entropy generation analysis on MHD flow with a binary mixture of ethylene glycol and water based silver-graphene hybrid nanoparticles in automotive cooling systems, *Int. J. Heat Technol.* 39 (6) (2021) 1781–1790.
- [9] W. Zhou, C. Ma, L. Yang, F. Luo, J. Liu, Regulation of thermo-fluid-solid coupling characteristics in high-speed spindle-bearing system for boring machine tool based on sintered-core heat pipes, *Int. Commun. Heat Mass Transfer* 157 (2024) 107717.
- [10] S.R.R. Reddy, H.T. Basha, P. Duraisamy, Entropy generation for peristaltic flow of gold-blood nanofluid driven by electrokinetic force in a microchannel, *Eur. Phys. J. Spec. Top.* (2022) 1–15.
- [11] S. Srinivas, A. Vijayalakshmi, A. Subramanyam Reddy, Flow and heat transfer of gold-blood nanofluid in a porous channel with moving/stationary walls, *J. Mech.* 33 (3) (2017) 395–404.
- [12] S.R.R. Reddy, P.B.A. Reddy, Bio-mathematical analysis for the stagnation point flow over a non-linear stretching surface with the second order velocity slip and titanium alloy nanoparticle, *Front. Heat Mass Transfer* 10 (13) (2018) 1–11.
- [13] F. Anene, J. Aiza, I. Zainol, A. Hanim, M.T. Suraya, Additively manufactured titanium alloys and effect of hydroxyapatite coating for biomedical applications: A review, *Proc. Inst. Mech. Eng. L* 234 (11) (2020) 1450–1460.
- [14] D.J. Fernandes, C.N. Elias, R.Z. Valiev, Properties and performance of ultrafine grained titanium for biomedical applications, *Mater. Res.* 18 (6) (2015) 1163–1175.
- [15] D.H. Teja, P. Muvvala, N.A.P. Nittala, D. Bandhu, M.I. Khan, K.K. Saxena, M.I. Khan, Comparative performance analysis of recuperative helium and supercritical CO₂ Brayton cycles for high-temperature energy systems, *Energy* 312 (2024) 133469.
- [16] Y.M. Chu, B.M. Shankaralingappa, B.J. Gireesha, F. Alzahrani, M.I. Khan, S.U. Khan, Combined impact of cattaneo-christov double diffusion and radiative heat flux on bio-convective flow of maxwell liquid configured by a stretched nano-material surface, *Appl. Math. Comput.* 419 (2022) 126883.
- [17] D. Li, H. Li, Electromagnetohydrodynamic (EMHD) flow of jeffrey fluid through parallel plate microchannels with surface charge-dependent asymmetric slip, *Phys. Lett. A* (2024) 130026.
- [18] S.R.R. Reddy, P.B.A. Reddy, Thermal radiation effect on unsteady three-dimensional MHD flow of micropolar fluid over a horizontal surface of a parabola of revolution, *Propuls. Power Res.* 11 (1) (2022) 129–142.
- [19] S.R.R. Reddy, P.B.A. Reddy, Numerical simulations of unsteady 3D MHD micropolar fluid flow over a slendering sheet, *J. Appl. Comput. Mech.* 7 (3) (2021) 1403–1412.
- [20] S.R.R. Reddy, P. Bala Anki Reddy, K. Bhattacharyya, Effect of nonlinear thermal radiation on 3D magneto slip flow of Eyring-Powell nanofluid flow over a slendering sheet with binary chemical reaction and Arrhenius activation energy, *Adv. Powder Technol.* 30 (12) (2019) 3203–3213.
- [21] L. Sun, G. Wang, C. Zhang, Experimental investigation of a novel high performance multi-walled carbon nano-polyvinylpyrrolidone/silicon-based shear thickening fluid damper, *J. Intell. Mater. Syst. Struct.* 35 (6) (2024) 661–672.
- [22] B. Mahanthesh, J. Mackolil, S.A. Shehzad, Statistical analysis of stagnation-point heat flow in williamson fluid with viscous dissipation and exponential heat source effects, *Heat Transfer* 49 (8) (2020) 4580–4591.
- [23] M. Nazeer, M.I. Khan, Y.M. Chu, S. Kadry, M.R. Eid, Mathematical modeling of multiphase flows of third-grade fluid with lubrication effects through an inclined channel: analytical treatment, *J. Dispers. Sci. Technol.* 43 (10) (2022) 1555–1567.
- [24] K. Govindarajulu, A.S. Reddy, Impacts of joule heating and viscous dissipation on MHD pulsatile flow of third grade nanofluid in a channel, *Proc. Inst. Mech. Eng. E* (2022) 1–12.
- [25] J. Zhu, S. Pan, H. Chai, P. Zhao, Y. Feng, Z. Cheng, J. Zhu, S. Pan, H. Chai, P. Zhao, Y. Feng, Z. Cheng, S. Zhang, W. Wang, Microfluidic impedance cytometry enabled one-step sample preparation for efficient single-cell mass spectrometry, *Small* (2024) 2310700.
- [26] T. Malathy, S. Srinivas, A.S. Reddy, Chemical reaction and radiation effects on MHD pulsatile flow of an Oldroyd-B fluid in a porous medium with slip and convective boundary conditions, *J. Porous Med.* 20 (4) (2017) 287–301.
- [27] M.A. Iqbal, Y. Wang, M.M. Miah, M.S. Osman, Study on Date-Jimbo-Kashiwara-Miwa equation with conformable derivative dependent on time parameter to find the exact dynamic wave solutions, *Fractal Fract.* 6 (4) (2022).
- [28] B. Guo, A. Raza, K. Al-Khaled, S.U. Khan, S. Farid, Y. Wang, M.I. Khan, M.Y. Malik, S. Saleem, Fractional-order simulations for heat and mass transfer analysis confined by elliptic inclined plate with slip effects: A comparative fractional analysis, *Case Stud. Therm. Eng.* 28 (2021) 101359.
- [29] I.M. Eldesoky, Slip effects on the unsteady MHD pulsatile blood flow through porous medium in an artery under the effect of body acceleration, *Int. J. Math. Math. Sci.* 2012 (2012) 1–26.
- [30] A.R. Bestman, Pulsatile flow in heated porous channel, *Int. J. Heat Mass Transfer* 25 (5) (1982) 675–682.
- [31] N. Datta, D.C. Dalal, S.K. Mishra, Unsteady heat transfer to pulsatile flow of a dusty viscous incompressible fluid in a channel, *Int. J. Heat Mass Transfer* 36 (7) (1993) 1783–1788.
- [32] G. Radhakrishnamacharya, M.K. Maiti, Heat transfer to pulsatile flow in a porous channel, *Int. J. Heat Mass Transfer* 20 (2) (1977) 171–173.
- [33] P.S. Reddy, P. Sreedevi, V.N. Reddy, Entropy generation and heat transfer analysis of magnetic nanofluid flow inside a square cavity filled with carbon nanotubes, *Chem. Thermodyn. Therm. Anal.* 6 (2022) 100045.
- [34] Y. Zhou, R. Duan, Leak-proof reversible thermochromic microcapsule phase change materials with high latent thermal storage for thermal management, *ACS Appl. Energy Mater.* 7 (14) (2024) 5944–5956.
- [35] S. Das, R.N. Jana, O.D. Makinde, MHD flow of Cu-Al₂O₃/water hybrid nanofluid in porous channel: Analysis of entropy generation, *Def. Diffus. Forum* 377 (2017) 42–61.
- [36] S. Saha, A. Kumar, A.N. Das, Generation of entropy of turbulent EG-water-Al₂O₃ hybrid nanofluid flow through a channel of rectangular cross-section, *Int. J. Ambient Energy* (2021) 1–11.
- [37] M.I. Khan, S. Kadry, Y. Chu, M. Waqas, Modeling and numerical analysis of nanoliquid (titanium oxide, graphene oxide) flow viscous fluid with second order velocity slip and entropy generation, *Chin. J. Chem. Eng.* 31 (2021) 17–25.
- [38] S.V. Buren, W. Polifke, Heat transfer in pulsating flow and its impact on temperature distribution and damping performance of acoustic resonators, in: *Future Space-Transport-System Components under High Thermal and Mechanical Loads: Results from the DFG Collaborative Research Center TRR40*, 2021, pp. 97–111.
- [39] P.B. Kumar, S. Srinivas, Pulsating flow of a non-Newtonian nanofluid in a porous channel with magnetic field, *Mater. Today: Proc.* 9 (2019) 320–332.
- [40] K. Govindarajulu, A.S. Reddy, Impacts of joule heating and viscous dissipation on MHD pulsatile flow of third grade nanofluid in a channel, *Proc. Inst. Mech. Eng. E* 236 (4) (2022) 1544–1555.
- [41] O.D. Makinde, T. Chinyoka, Numerical study of unsteady hydromagnetic generalized Couette flow of a reactive third-grade fluid with asymmetric convective cooling, *Comput. Math. Appl.* 61 (4) (2011) 1167–1179.